

Notes: Li & Mann (RFS, 2025)

Digital Tokens and Platform Building

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1 Big picture

- **Question.** What economic role can a blockchain-based digital token play when an entrepreneur launches a peer-to-peer platform?
- **Core answer.** A platform-specific token can solve the platform’s coordination problem. The token sale lets users credibly communicate their intention to participate before the platform launches.
- **Why this is not standard financing.** The token in the model pays no dividend, is not repurchased by the entrepreneur, and has no use outside the platform. Its value comes from coordinating platform users, not from a financial cash flow.
- **Coordination problem.** Users want to join the platform only if they expect other users to join. This strategic complementarity creates multiple equilibria, including a no-trade equilibrium.
- **Mechanism.** Buying an intrinsically useless token is a costly, publicly observable action. Under forward induction, other users infer that the buyer must intend to use the platform; otherwise the purchase would be irrational.
- **Main result.** In the main symmetric-platform model, any forward-induction outcome is a success outcome: all tokens are sold and the maximum volume of platform trade occurs at every platform date.
- **Key distinction.** The token does not subsidize participation, prepay the platform participation cost, or commit the buyer to participate. It only changes beliefs about future participation.
- **Why intrinsic uselessness helps.** If the token had dividends, general monetary usefulness, or other independent value, buying it would no longer clearly signal intent to use the platform.
- **Extensions.** The model adds non-user financial investors, late adopters, and private information about platform value. The same forward-induction logic delivers platform success and can make the token auction aggregate dispersed information.
- **Policy implication.** Some tokens can have a utility-token role distinct from securities. Regulation should preserve the coordination function while protecting investors from abuse.

2 Data and measurement: what counts as “data” in this theory paper

2.1 No empirical dataset; the paper uses institutional examples

- This is a theory paper. It does not estimate regressions or construct a dataset.
- Its motivating evidence is institutional: many blockchain projects launched platforms by selling or distributing native tokens before platform use began.
- The examples establish the economic environment the model targets: peer-to-peer platforms with network externalities and a native medium of exchange.

Interpretation: The paper’s “data section” is best read as a mapping from real token projects to model primitives. The examples justify why it is reasonable to model tokens as platform-specific media of exchange sold before launch.

2.2 Platform examples used to motivate the model

- **Cross-border payments and remittances.** Ripple/XRP and Stellar/XLM use a native token to facilitate settlement or bridge-currency functionality.

- **Storage, computing, and file sharing.** Filecoin, Arweave, Sia, Golem, and BitTorrent use tokens to connect users who supply and demand decentralized resources.
- **Social networks and content platforms.** Kin, TON, Steemit, and Audius use tokens to reward activity, coordinate users, and support platform governance or exchange.
- **Blockchains.** Ethereum, Mastercoin, and Algorand are examples where a blockchain or smart-contract platform launches with an internal medium of exchange.

Interpretation: All examples share a common tension: a platform has value only if enough other users participate. A token sale can make participation beliefs self-confirming in the good direction.

2.3 The central measured object: observable token acquisition

- The model requires a costly action before the platform launches.
- The action must be publicly observable.
- Blockchain transparency makes token acquisition especially visible.
- The token must be usable only on the platform, so acquisition signals intended platform participation.

Interpretation: The blockchain is not valuable here because it mechanically creates cash flows. It is valuable because it makes the costly signal transparent to dispersed users.

3 Model structure and theoretical specifications

3.1 One-line mechanism

public token sale $\Rightarrow$ costly observable token acquisition $\Rightarrow$ belief that purchasers will participate $\Rightarrow$ platform co

Interpretation: The token sale is a communication technology. It turns a future participation intention into an observable pre-launch action.

3.2 Trivial two-user platform model

Two users, A and B , choose actions y or n at the platform date.

- Choosing n gives payoff 0.
- Choosing y costs $u > 0$.
- If both choose y , each receives an additional payoff $s > u$.

The one-shot game has two natural equilibria:

$$(y, y) \text{ success, } (n, n) \text{ failure.}$$

Now add a token-sale stage. At $t = 0$, user A can choose “in” or “out.” If A chooses in, she pays

$$P \in (0, s - u),$$

and only then can the platform game be played at $t = 1$.

Interpretation: Without the token sale, failure is self-confirming. With the token sale, A 's choice to pay P is hard to rationalize unless she intends to choose y later.

3.3 Forward induction logic

The paper follows Govindan and Wilson's framework, built on Reny's formalization of forward induction. The key concepts are:

- **Weakly sequential best reply:** a pure strategy that is optimal at every node it does not exclude.
- **Weakly sequential equilibrium (WSE):** strategies and consistent beliefs such that each pure strategy in support is a weakly sequential best reply.
- **Relevant strategy:** a strategy that can be rationalized as a best reply to some equilibrium with the same outcome.
- **Forward induction outcome:** a WSE outcome where beliefs at relevant nodes are restricted to relevant strategies.

Interpretation: If a player deviates by making a costly observable move, others should not immediately conclude irrationality. They should ask what rational continuation could justify the costly move.

3.4 Proposition 1: token sale eliminates failure in the trivial model

The paper's first formal result is:

The only forward-induction outcome is the success outcome.

Interpretation: If A pays for the token, B should infer that A intends to use it. Then B wants to participate, and A anticipates this. The token sale transforms the equilibrium-selection problem.

3.5 Main symmetric platform model

There are $N \geq 2$ potential users and N tokens. The entrepreneur sells tokens before platform launch.

- Token sale runs from dates $t = 1$ to $t = N$.
- At each sale date, any user without a token chooses whether to purchase one for price P .
- If no tokens are sold, the game ends with zero payoff.
- If any tokens are sold, platform operation starts at date $N + 1$.

On the platform:

- a user can request the service at utility cost u ;
- a user can offer the service at utility cost u ;
- a completed transaction transfers a token from demander to supplier;
- the demander obtains benefit s ;
- the supplier pays service cost c ;
- each user can hold at most one token.

Maximum trade means every user both supplies and obtains the service each platform date, earning flow utility

$$s - c - 2u.$$

Interpretation: The token is modeled as the exclusive medium of exchange. A user needs it to obtain the service, and a supplier receives it in exchange for supplying the service.

3.6 Key assumptions in the symmetric model

The paper assumes:

$$\rho \in (0, 1),$$

where ρ is the discount factor between platform dates, and

$$\rho(s - u) - (c + u) > 0.$$

This condition means it is valuable to acquire a token on the platform and spend it in the next date.

For simplicity, the main model sets

$$P = c + u,$$

so the cost of acquiring a token in the sale equals the cost of acquiring it through platform trade. Appendix F relaxes this to $P \leq c + u$ with a richer equilibrium concept.

Interpretation: These assumptions make platform interaction socially valuable and make token acquisition meaningful without giving the token independent financial value.

3.7 Failure and success outcomes

- **Failure:** no tokens are sold and no trade occurs.
- **Success:** all tokens are sold and maximum trade occurs every platform date.

If success occurs, each user earns lifetime utility

$$\frac{\rho}{1 - \rho}(s - c - 2u) - P.$$

Interpretation: The theorem characterizes outcomes, not one particular strategy profile. Many equilibria may differ in timing and off-path behavior, but all forward-induction outcomes have the same economic content: full adoption.

3.8 Weakly monotonic equilibrium

The repeated platform game is simplified by restricting strategies to depend only on:

- the user's token-holding status $X_i(h) \in \{0, 1\}$;
- the total number of tokens held by users at node $M(h)$.

A pure strategy is weakly monotonic if choosing y at a node implies also choosing y at any comparable node with the same token status and weakly more tokens sold:

$$X_i(h') = X_i(h), \quad M(h') \geq M(h).$$

Interpretation: More tokens sold should not make coordination less likely. This rules out pathological equilibria where users abandon the platform precisely because adoption is higher.

3.9 Theorem 1: symmetric platform

The paper's main theorem is:

Any forward-induction outcome is a success outcome.

Thus every player buys a token during the sale, and every player both supplies and obtains the service at every date on the platform.

Interpretation: The theorem says the token sale eliminates the no-trade equilibrium. The effect comes from beliefs, not from subsidies or commitment.

3.10 Extension: investors and late adopters

The paper then adds financial investors and late adopters.

- The original users become **early adopters**, with N_1 tokens.
- The entrepreneur also creates N_2 extra tokens.
- At $t = 0$, at least $2N_2$ investors bid for these extra tokens.
- Later, at date $\bar{T}$, a group of late adopters arrives.
- Investors resell the extra tokens to late adopters.

The paper assumes a long enough time gap between launch and late-adopter arrival:

$$\frac{\rho^{\bar{T}-N_1}}{1-\rho}(s-c-2u) < P.$$

Interpretation: The condition prevents early adopters from profitably buying tokens only to wait and resell to late adopters. It keeps the early token sale focused on coordination, while investors intermediate later demand.

3.11 Theorem 2: late adopters

Theorem 2 states that any forward-induction outcome is a success outcome. In equilibrium:

- all N_2 extra tokens are sold to investors initially;
- investors later sell them to late adopters;
- the initial investor price and later resale price are both

$$\frac{s-c-2u}{1-\rho}.$$

Interpretation: Investors can rationally buy tokens even though they never use the platform. They buy future access rights and resell them once the early adopters have coordinated.

3.12 Extension: investors with private signals

Now platform value s is unknown at $t = 0$.

- s is drawn from a known distribution on $[0, \bar{s}]$.

- Investors observe private signals v_i about s .
- The signals satisfy a monotone likelihood ratio property.
- At $t = 1$, s is revealed and the rest of the model proceeds as before.

Define users' willingness to pay for a token as

$$V \equiv \max \left\{ 0, \frac{s - c - 2u}{1 - \rho} \right\}.$$

Interpretation: Investor trading is speculative in this extension, but the object being priced is still future platform access, not a dividend claim.

3.13 Theorem 3: information aggregation

Theorem 3 states that any forward-induction outcome is a success outcome and the auction price aggregates investor signals:

$$(P^{auc} - V) \xrightarrow{p} 0 \quad \text{as } N \rightarrow \infty.$$

Interpretation: The token auction can reveal information about future platform value even though the token has no intrinsic value. The market price reflects expected future user demand for platform access.

4 Identification, equilibrium selection, and credibility of the mechanism

4.1 What identifies the mechanism in a theory paper

There is no empirical identification strategy. The mechanism is identified logically by comparing environments:

- platform without token sale;
- platform with token sale but without forward induction;
- platform with both token sale and forward induction.

Interpretation: The token sale matters only when users reason from costly past actions to future intentions.

4.2 Why no-trade survives without the token sale

If tokens are distributed for free, users take no costly pre-launch action. The no-trade equilibrium remains:

$$\text{everyone expects nonparticipation} \Rightarrow \text{nonparticipation is optimal.}$$

Interpretation: Free distribution does not create the same signal. The key is not token ownership by itself, but costly observable token acquisition.

4.3 Why no-trade survives without forward induction

Even with a token sale, a weakly monotonic equilibrium can support failure if users believe token purchasers still will not use the platform.

Interpretation: A standard solution concept may treat costly purchases as off-path mistakes. Forward induction instead asks what rational continuation could justify them.

4.4 Why multi-date token sales matter

The proof of Theorem 1 uses the multi-date token sale to rule out partial-participation equilibria.

- With a single sale date, users could randomize and coordinate only if all tokens are sold.
- With multiple sale dates, if reaching full token adoption would induce success, then each remaining user has an incentive to move the economy closer to that state.
- This unraveling logic rules out partial adoption.

Interpretation: The multi-date structure is not merely cosmetic. It helps extend forward-induction reasoning from two players to many users.

4.5 Why the assumptions are economically important

- **Exclusive medium of exchange.** Only token holders can benefit from platform trade.
- **Observable purchase.** Others can see that the user paid a cost to acquire the token.
- **Intrinsic uselessness.** Token purchase cannot be easily explained by dividends, general liquidity, or consumption value.
- **Platform gains from trade.** Successful interaction must be valuable enough to justify costly adoption.

Interpretation: Each assumption rules out an alternative explanation for token acquisition. This is what makes the purchase a credible signal.

4.6 What the model cannot fully prove

- It does not empirically estimate whether actual token purchasers use forward induction.
- It does not model fraud, hype, manipulation, or bubbles directly.
- It abstracts from many token-market frictions, such as endogenous secondary-market liquidity and heterogeneous user tastes.
- It assumes observable purchases and common understanding of platform rules.
- It treats the entrepreneur's commitment to token exclusivity as credible.

Interpretation: The model provides a clean economic rationale for utility tokens, but not a universal defense of all token sales.

5 Paper roadmap: section-by-section detailed notes

5.1 Introduction

- The paper begins from the puzzle of ICOs and token launches: why create a custom digital token instead of using debt, equity, or a pre-existing currency?
- The answer is that token sales can solve coordination failure on peer-to-peer platforms.

- The token’s blockchain transparency makes purchase observable; its platform-specific nature makes purchase informative.
- The authors emphasize that the token can be valuable even if it pays no dividend and has no use outside the platform.

5.2 Section 1: coordination and tokens in platform projects

- The paper surveys real token projects whose platform value depends on user coordination.
- Cross-border payments, decentralized storage, content platforms, and blockchains all require enough users or counterparties to be useful.
- Many projects sold tokens for cash or distributed them in exchange for costly effort before launch.

5.3 Section 2: trivial two-user platform model

- A simple two-player coordination game has both success and failure equilibria.
- Adding a pre-play token purchase changes the interpretation of later actions.
- Under forward induction, the only outcome is success.
- The token sale works even though the purchaser still pays the platform participation cost later.

5.4 Section 3: many symmetric users

- The paper builds an N -user repeated platform model with a token sale and platform operation.
- Users can request and offer the service; tokens move from demanders to suppliers.
- Success means all tokens are sold and maximum trade occurs forever.
- Theorem 1 generalizes the simple two-user logic to many players and an infinite horizon.

5.5 Section 4.1: investors with perfect information

- The model adds investors who buy extra tokens before launch but do not use the platform.
- Investors later resell to late adopters once the platform is already active.
- The forward-induction outcome remains success.
- Investors are intermediaries between the entrepreneur and future users.

5.6 Section 4.2: investors with private signals

- Investors receive private signals about platform value.
- The initial token auction aggregates these signals.
- The auction price converges to users’ willingness to pay for platform access.
- This gives token sales an information-revelation role in addition to the coordination role.

5.7 Section 5: discussion and implications

- Tokens are most effective as coordination devices when they lack other sources of value.
- This creates a tension with efforts to make tokens look like securities or general-purpose currencies.
- The paper discusses implications for the Howey test and for token regulation.
- It also compares token sales to membership sales and discusses efficiency conditions.

5.8 Section 6: conclusion

- The conclusion restates the paper’s central claim: tokens can prevent inefficient coordination failure.
- The authors emphasize transparency, platform specificity, and lack of outside value as sources of coordination power.
- They also acknowledge simplifying assumptions and point to richer models of token holdings and platform shocks.

6 Original equation panels

6.1 Platform examples and the trivial token-sale model

Read:

- The examples connect the model to real token-based platforms: remittances, storage, social networks, and blockchains.
- The trivial model shows why costly token acquisition can make the success equilibrium focal.
- The important comparison is not token versus no token mechanically, but costly observable token purchase versus no costly observable action.

Economic message: Token sales can be interpreted as pre-launch coordination devices for platforms facing strong network externalities.

as the medium of exchange, although they vary in the details of exactly how platform users interact with each other. Some of them are best described by our main setup in [Section 3](#), where the platform is completely symmetric with each user being able to both supply and demand services at every date. Others more closely match the setup in [Appendix D](#), with users alternating between demanding and supplying the service, or the setup in [Appendix E](#), which fixes each user's type and introduces an exchange where service suppliers can sell tokens back to service demanders. As we will show, however, our main result holds in all three setups with essentially the same proof strategy.

Cross-border payments and remittances. Cross-border payments traditionally involve financial institutions that must maintain pre-funded accounts in various currencies, which is a costly and inefficient use of capital. In principle, a platform system can provide clearance and settlement services that would economize on this capital, much like a traditional exchange. Several projects have attempted to implement this idea using blockchain technology.

Ripple (more precisely RippleNet) is one prominent example. It facilitates cross-border transactions using its native token XRP as the bridge currency. XRP tokens were initially distributed as bounties to users for improving code in the system, then eventually sold more broadly to individuals via anonymous exchange transactions, and to institutional investors through direct sales. Ripple is also noteworthy for having had some success defending itself against the SEC, as we mentioned in the introduction and discuss further in [Section 5](#).³

Another similar project is Stellar, with its native token XLM, which has partnered with both IBM and MoneyGram to facilitate cross-border payments and remittances.

Storage, computing, and file sharing. Many platforms propose to connect people to share idle computing resources. These projects face the obvious coordination problem that it is difficult to attract users to the network when others are not already present.

The best-known example in this space is Filecoin, which aimed to provide a peer-to-peer platform through which users would effectively rent out their spare capacity. Those users received a native digital token (called filecoin) as payment. The project launched with a token sale in August 2017 that attracted approximately \$200 million in proceeds. [Howell, Niessner, and Yermack \(2020\)](#) provide a detailed case study of this project.

Other storage networks following a similar setup are Arweave (with its native token called AR), and Sia (Siacoin). Similar ideas are implemented for sharing computer power by Golem (GLM), and file-sharing by BitTorrent

(a) Motivating platform examples.

n results in a payoff of zero. Choosing y results in a cost of $u > 0$. Then, if and only if both players chose y , they each receive a further payoff $s > u$. Clearly, there is a “success” equilibrium in which both players choose y , and a “failure” equilibrium in which both choose n . For our purposes, interpret this as the simplest possible model of a platform.

Now imagine this coordination game is played at $t=1$, but only if A paid a cost at $t=0$. That is, assume A chooses between “in” and “out” at $t=0$. If she chooses “out” the game ends. If she chooses “in,” she pays a cost $P \in (0, s-u)$, and we proceed to the coordination game. We can interpret date 0 as the sale of a token for later use on the platform.

The idea of forward induction is that the “failure” outcome now seems unreasonable. Suppose A pays the cost P at date 0. Intuitively, B should reason that this action could not make sense if A then plans to choose n at date 1. It seems that B should instead expect A to play y at $t=1$ in this case, and then B should rationally play y as well. In turn, A should anticipate this reasoning and choose (in, y), leading to coordination success.

The literature has proposed several formalizations of this argument. We follow [Govindan and Wilson \(2009\)](#), hereafter GW2009, who in turn build on [Reny \(1992\)](#). In their approach, equilibrium specifies each player's beliefs about others' strategies, and forward induction restricts those beliefs to maintain the common belief of rationality wherever possible.

Formally, consider a finite game of complete information and perfect recall, represented by a game tree, set of players and actions, payoffs defined over terminal nodes, and outcomes defined as distributions over those terminal nodes. GW2009 model forward induction as a refinement of weakly sequential equilibrium (WSE). WSE comprises two ingredients:

(b) Trivial two-user platform model.

6.2 Forward induction definitions and Proposition 1

Read:

- The definitions formalize how players should interpret costly deviations.
- Proposition 1 states that the only forward-induction outcome in the simple model is success.
- The logic is that (in, y) is rationalizable after A pays the cost, whereas (in, n) is not.

Economic message: The formal equilibrium refinement gives discipline to the intuitive statement: buying a useless token means the buyer plans to use it.

Definition 1. A pure strategy s_i for a player i is a weakly sequential best reply to her beliefs μ_i if s_i , at every node it does not exclude, prescribes an optimal continuation given μ_i .

Definition 2. A weakly sequential equilibrium is a pair (b, μ) of a strategy profile b , and a consistent beliefs profile μ , such that for each player i , each pure strategy in the support of the equivalent normal-form strategy is a weakly sequential best reply to μ_i .

As the name suggests, WSE is slightly weaker than sequential equilibrium, in that it does not restrict actions at nodes following one's own deviation ("does not exclude"). [Reny \(1992\)](#) and [GW2009](#) discuss why this is the appropriate concept when studying forward induction.

Next, forward induction refines the set of WSE by requiring that, wherever possible, players' beliefs maintain the assumption that others are rational. To be precise, each player i believes where possible that each other player j is playing a best response to some equilibrium that may differ from the one i thinks is being played, but at least has the same outcome.

Note that this restriction is automatically satisfied along the equilibrium path, and only bites when j deviates to an action that (i believes) is strictly inferior to j 's equilibrium strategy. In this case, rather than jump to the conclusion that j is irrational, i asks if j 's action could instead be explained as a disagreement about what equilibrium strategies specify off the equilibrium path. If so, then i assumes that j will continue to act rationally subject to her belief. This in turn may influence how i responds.

Definition 3. A pure strategy for a player i is a relevant strategy to a WSE outcome if it is a weakly sequential best reply for i to some WSE with that outcome.

Definition 4. A node is a relevant node to a WSE outcome if it is reached by some profile of strategies that are relevant to that outcome.

Definition 5. A forward induction outcome is the outcome of a WSE in

(a) Weakly sequential equilibrium and forward induction definitions.

A 's deviation, while (in, n) cannot. Forward induction then requires that if B gets the chance to move, he believes A is playing (in, y) . B should then choose y , and A should anticipate this reasoning and play (in, y) .

Proposition 1 (Trivial model). The only forward induction outcome is the success outcome.

The point is that an entrepreneur attempting to launch the platform at date $t = 1$ could overcome potential coordination problems between the two users by holding a token sale at $t = 0$. To do so, the entrepreneur must commit not to allow A to use the platform unless she paid the cost at $t = 0$. However, nothing else changes about the game played at $t = 1$. In particular, A does not commit herself to choosing y at that date, nor does she avoid or prepay the participation cost u from doing so. This simple model illustrates our central argument: token sales can achieve coordination by exploiting users' forward induction reasoning.

We note that forward induction itself is based on realistic assumptions about how people think. For example, [Battigalli and Siniscalchi \(2002\)](#) derive forward induction from an explicit epistemic framework, and [van Huyck, Battalio, and Beil \(1993\)](#) demonstrate its real-life power in an experiment with a striking resemblance to our model.⁵ Thus, mechanisms based on this reasoning potentially deserve more attention in the literature.

Other approaches to achieving coordination often involve subsidizing early users for their participation. Our token-based approach exploiting forward induction is very different. User A pays a strictly positive cost for the right to participate in the platform game with B at $t = 1$, and neither expects nor receives any compensation from the entrepreneur for doing this. This observation will

(b) Proposition 1 and interpretation.

6.3 Main platform setup and outcome definitions

Read:

- The main model introduces N users, N tokens, a multi-date token sale, and infinite-horizon platform operation.
- The platform clears as many transactions as possible using tokens as the medium of exchange.
- Success is defined as all tokens sold and maximum trade every date.

Economic message: The token's role is endogenous to platform exchange: it is needed to request the service and is earned by supplying the service.

We now extend the previous model in two ways. First, we allow an arbitrary number of tokens and potential platform users. Second, we explicitly model users' interactions by assuming that they provide each other a service on the platform, with their transactions facilitated by the tokens they purchased beforehand.

We derive our main result using a few different specifications of the platform, since there is no single model that can describe every variety of token-based project in practice. In the main paper, the platform is completely symmetric, with each user able both to supply and demand the service at every date. [Appendix D](#) presents an alternative setup with users alternating between demanding and supplying the service. [Appendix E](#) fixes each user's type, and introduces an exchange where the service suppliers can sell the tokens back to the demanders. Each case leads to the same main result with essentially the same proof strategy. The main paper focuses on the symmetric model primarily because it is the simplest.

3.1. Setup

An entrepreneur launches a platform that has $N \geq 2$ potential users. Before the platform launch, the entrepreneur offers N tokens in a sale lasting from dates $t = 1$ through $t = N$. At each of these dates, each user who does not yet hold a token chooses $\{y, n\}$ whether to purchase one for price P . When the sale ends at date N , if no tokens were sold, then the game reaches a terminal node with zero payoff for everyone. If any tokens were sold, then we proceed to an infinitely repeated game of platform operation starting from $t = N + 1$.

At each platform node, users can choose whether to request the service and whether to offer it. Either choice bears a utility cost of u . Given users' choices, the platform then clears as many transactions as possible. Each transaction requires the transfer of a token from a user who owns it and requests the service, to a user who offers the service, with the supplier paying a utility cost of c and the receiver obtaining a utility benefit of s . If demand and supply are imbalanced at any date, then the platform chooses counterparties randomly with equal probability from the side with more participants.

For tractability, we assume that users can only hold one token at a time, similar to storage constraints in many models of money (e.g., [Kiyotaki and Wright 1989](#); [Trejos and Wright 1995](#)). This means they can only purchase once in the sale. It also means that the only way they can supply the service while holding a token is if they simultaneously spend that token.

We allow users to both request and offer the service at the same time, which costs them $2u$. If matched for both transactions, then they can supply and receive the service at the same time as well.⁷ We also allow users to attempt

actions that will certainly fail, such as requesting the service when they do not hold a token, or offering the service when they already hold a token and do not try to spend it. In such cases they simply pay the cost u .

Because platform operation is an infinitely repeated game, an outcome is now a distribution over paths through the game tree. We let "failure" denote the outcome in which no tokens are sold and no trade occurs. We let "success" denote any outcome in which all tokens are sold and the maximum amount of trade occurs every date on the platform (that is, each user both supplies and obtains the service every date, earning flow utility of $s - c - 2u$).⁸

We assume a common discount rate $\rho \in (0, 1)$ between platform dates. We assume $\rho(s - u) - (c + u) > 0$, so that it is worth acquiring the token on the platform and spending it the next date. For simplicity, we assume no discounting during the token sale, but we could change this with no important effect. Finally, also for simplicity, we set $P = c + u$ so that acquiring the token in the sale or on the platform incurs the same utility cost. [Appendix F](#) shows that we can relax this to $P \leq c + u$ with a slightly richer equilibrium concept.

3.2. Equilibrium concept

In analyzing our repeated game, we strengthen the WSE concept as follows: First, we extend WSE to an infinite-horizon game through obvious changes to the definitions.⁹ Next, we assume that each pure strategy s_i in each player i 's equilibrium strategy conditions at any node h only on i 's token holding status $X_i(h) \in \{0, 1\}$, and on the total number of tokens owned by players at that node, $M(h)$. Among other things, this assumption rules out time-dependent strategies. Finally, we assume that players are not less likely to choose y for larger M . We label the resulting concept weakly monotonic equilibrium (WME).

Definition 6. Take any node h , and any pure strategy s_i for player i . We say that s_i is weakly monotonic if, whenever s_i specifies y for a given action at a node h , then it also specifies y for the same action at any h' where $X_i(h') = X_i(h)$ and $M(h') \geq M(h)$.

Definition 7. A weakly monotonic equilibrium is a weakly sequential equilibrium in which the strategy profile is restricted to weakly monotonic strategies.

⁸ In [Section 2](#), "success" was a single outcome, but here it is strictly speaking a category of outcomes. This is because there are many paths through the game tree along which all tokens are sold, differing only in the timing of those transactions. Any distribution over them is a success outcome.

⁹ Recall that the GW2009 framework focused on finite games, and relied on the equivalence of behavior and mixed strategies for finite games of perfect recall that was established by [Kuhn \(1953\)](#). In copying their definitions to an infinite game, we invoke the extension of Kuhn's result to infinite games by [Aumann \(1964\)](#).

⁷ For example, think of a social media platform where users provide each other with user-generated creative content like writing or music, or a decentralized storage platform where users provide each other with specific records. [Appendices D and E](#) present alternative specifications that may better fit other platforms.

(a) Symmetric platform setup.

(b) Failure, success, and weak monotonicity.

6.4 Theorem 1 and the late-adopter extension

Read:

- Theorem 1 states that the symmetric platform's only forward-induction outcome is success.
- The late-adopter extension shows that non-user investors can buy tokens early and resell them to future users.
- This explains why some early token purchasers may look like intermediaries rather than platform users.

Economic message: The model separates the coordination role of the early token sale from the resale and access-pricing role of later token markets.

We assume that there is a sufficiently long gap $\bar{T} - N_1$ between the platform launch (at date N_1) and the arrival of the late adopters (at $\bar{T}$) such that

$$\frac{\rho^{\bar{T}-N_1}}{1-\rho}(s-c-2u) < P. \quad (1)$$

Theorem 1 (Symmetric platform). Any forward induction outcome is a success outcome.

That is, in any equilibrium outcome consistent with forward induction, every player purchases a token during the sale, and each player both supplies and obtains the service every date on the platform. Each player's utility flow each date on the platform is $s - c - 2u$, for a lifetime utility of $\frac{\rho}{1-\rho}(s - c - 2u) - P$.

Hence, an entrepreneur seeking to coordinate a large user base on a platform could do so by creating digital tokens to be the medium of exchange, and then selling them to the users before the platform launch. This works even though the tokens create no commitment, do not subsidize early adopters, and do not avoid the participation cost of the platform.

As before, tokens in this model have no intrinsic value. They pay no dividends, do not appreciate in price, and will never be repurchased. In fact, the token's effectiveness in coordinating users arises not despite but because of its uselessness for any other purpose. By purchasing such a token, a user visibly pays a cost that cannot be recouped unless she participates on the platform. This value would actually be undermined if the token offered other sources of utility. We expand on this point in [Section 5](#) at the end of the paper.

As in [Section 2](#), the sale proceeds need not be used by the entrepreneur for any specific purpose, and the "purchase" of a token need not even represent a monetary payment to the entrepreneur at all. Again, it is sufficient that tokens are provided in exchange for a user incurring some observable cost, whether in monetary or utility terms.

Note that [Theorem 1](#) characterizes equilibrium outcomes, not specific

(a) Theorem 1: symmetric platform success.

This simplifies the analysis by implying that an early adopter cannot benefit from purchasing a token, then waiting to spend it until the late adopters arrive.

As before, "failure" refers to an outcome in which no tokens are sold and no trade occurs. "Success" now means an outcome that has N_1 transactions occurring at each platform through $\bar{T}$, and $N_1 + N_2$ transactions occurring at each date from $\bar{T} + 1$ onward.

Our equilibrium concept for this model is WME, as defined in [Section 3](#), with the addition of the obvious individual rationality requirements for investors and late adopters.

Theorem 2. (Late adopters). Any forward induction outcome is a success outcome.

In any forward induction outcome, all N_2 extra tokens are sold to investors initially for a price of $\frac{s-c-2u}{1-\rho}$, then sold to the late adopters for the same price. This matches the observation that platform tokens are often sold very early to purchasers who more resemble intermediaries than users. Indeed, the number of tokens sold in this way could be much larger than the number sold to the early adopters.

Intuitively, the entrepreneur still uses the token sale described in [Section 3](#) to coordinate the users who are present on the platform at its earliest date. They initiate the platform's successful equilibrium, and are rewarded by paying a lower token price. The entrepreneur can then (via the investors) charge the full price to other users who will join after observing the successful operation of the platform. In fact, as long as investors are convinced that the entrepreneur will

(b) Investors and late adopters.

6.5 Information aggregation and policy implications

Read:

- The private-signal extension makes investors speculators with information about platform value.
- The token auction price aggregates their signals about future user willingness to pay.
- The policy discussion highlights the tension between tokens as utility/coordination devices and tokens as securities.

Economic message: Token prices can reveal information about platform value even when the token is not a claim on cash flows.

4.2. Investors with private signals

We now assume that, at $t=0$, the value of the platform is not yet known. Instead, s is distributed on $[0, \bar{s}]$ according to a commonly known density, with $\rho(\bar{s} - u) - (c + u) > 0$. That is, for sufficiently high (low) s the platform is (is not) socially valuable.

The investors each get a private signal v_i about s at date $t=0$, drawn from a conditional distribution with a density function satisfying the monotone likelihood ratio property. Then, at $t=1$, s is revealed, and the rest of the model proceeds as in [Section 4.1](#). Because the investors now trade on information and face risk, they can be interpreted as speculators.

In this model, we define a success outcome as matching the earlier definition (all tokens sold and maximum volume of trade) whenever the realization of s is high enough that trade is socially valuable, and as matching the failure outcome otherwise.

As before, success is the only outcome consistent with forward induction. The new result of this section is that P_{auc} , the price realized in the initial auction, now aggregates the investors' signals to provide an estimate of $V \equiv \max\left(0, \frac{s - c - 2u}{1 - \rho}\right)$, the users' willingness to pay for the token (incorporating their real option to ignore it).

Theorem 3 (Information aggregation). Any forward induction outcome is a success outcome. The token auction aggregates investors' signals in the sense of [Kremer \(2002\)](#): $(P_{\text{auc}} - V) \rightarrow_p 0$ as $N \rightarrow \infty$, where the limit is across a sequence of independent copies of the model.

Thus, the token auction aggregates dispersed information among investors. This result may seem surprising, as the tokens being auctioned pay no dividends, cannot be consumed, and will only be in future demand if a different set of agents coordinates their actions on a platform that has not been launched yet. However, investors know that in equilibrium the token sale process will generate coordination success (whenever this is efficient), and this will create demand for the tokens, with users willing to pay up to the present value of an infinite horizon of trading on the platform.

(a) Theorem 3: information aggregation.

sale. In this case the funds clearly serve no financing purpose and, arguably, might not legally constitute an investment.¹⁴

However, questions about existing law are ultimately not our focus and must be left to regulators and courts. A more constructive contribution of our model is that, by highlighting the economic role of token sales, it may suggest ways that existing frameworks could be modified to preserve that role while still protecting the investor community.

For example, in [Section 4](#) many tokens are sold initially to financial investors, not users. A flexible framework for tokens could limit these early sales to accredited investors, but allow an exemption for targeted sales or reward-based distributions that resemble our token sale, while also allowing the accredited-investor restriction to expire once the platform is operational so that non-users can sell tokens on to users. This hybrid entity would arguably protect the interests of all parties, but does not seem to fit into current legal frameworks.

We offer this as just one example, and we caution again that our discussion of current law is not expert opinion. The main point is that, by offering a coherent economic explanation of tokens, our model can shed light on future discussions of what the optimal legal framework should be, in order to promote both efficiency and market integrity.

Money. While not primarily a model of money, our framework clearly has some connections to that literature that offer a useful perspective on our results.

Our token resembles money, being intrinsically worthless except as a medium of exchange. A standard feature of money models is the existence of an inefficient no-trade equilibrium, as described in [Kiyotaki and Wright \(1989, p. 943\)](#): "If no one believes in fiat money, then it cannot get off the ground [...] The value of any medium of exchange, and especially fiat money, ultimately depends at least partially on faith." We suggest that forward induction reasoning can create these beliefs and faith endogenously, if agents are given the opportunity to adopt money early on in a costly and visible way.

The requirement to use tokens for trade on the platform resembles a standard cash-in-advance constraint (e.g., [Lucas and Stokey 1987](#)). However, to our knowledge there is no prior model that suggests this approach as a way to harness forward induction and encourage money adoption. The typical cash-in-advance model has no explicit coordination problem, and therefore no inefficient equilibrium, even without forward induction.

Comparison with a membership model. An alternative approach might be for the entrepreneur to sell membership in the platform, then simply allow fiat currency as the medium of exchange. We do not know of any formal model of

¹⁴ The idea may sound far-fetched, but we do observe equivalent behavior in blockchain projects. For example, Mastercoin's token sale, often referred to as the first ICO, sent all its proceeds to a verifiably unspendable address, an action known as "burning" in the blockchain world. Interestingly, [Ben-Porath and Dekel \(1992\)](#) also motivate forward induction by describing agents "burning" money.

(b) Policy, money, membership, and efficiency discussion.

6.6 Appendix logic: eliminating non-success equilibria and generalizing price

Read:

- Proposition 2 shows that every weakly monotonic equilibrium outcome other than failure is success.
- This appendix result helps the proof focus on ruling out failure.
- Appendix F generalizes token pricing from $P = c + u$ to $P \leq c + u$ by allowing strategies to condition on recent trade volume $\hat{M}$.

Economic message: The appendix clarifies which assumptions are technical and which are economic. The core force is costly observable acquisition plus forward-induction reasoning.

— implies that the outcome of the experiment must be success.

Lemma 3. Suppose a pure strategy for a user i is weakly monotonic, relevant to the failure outcome, and specifies y at some token-sale node. Then, it specifies both to demand and to supply service at any platform node for which $M=N$.

Proof. Clearly the strategy must specify spending at some platform node or it would be strictly dominated by doing nothing at all. Then, by weak monotonicity it must also specify spending at any platform node with $M=N$. It cannot exclude such nodes if it specifies y during the token sale, so this spending must be part of a weakly sequential best reply in the continuation from such nodes. But, if the strategy does not also specify supplying the service at such nodes, then there is no chance of being chosen for trade, in which case the decision to spend would not be rational. So the strategy must also specify supplying the service at such nodes. ■

Proof of Theorem 1: By Proposition 2, we only need to prove that the failure outcome is not consistent with forward induction. Suppose by contradiction that there is a WME (b, μ) that specifies relevant strategies at all relevant nodes, whose outcome is failure. By Lemma 1, there is at least one relevant platform node with $M=N$, so equilibrium must specify relevant strategies for all users there. Then, by Lemma 3, all users must play (y, y) under b at any node with $M=N$. Then, by Lemma 2, the outcome of b must be success, a contradiction. ■

Appendix C Proofs for Section 4 (Adding Investors)

Proof of Theorem 2: We extend the steps in the proof of Theorem 1 to this setting.

First we extend Proposition 2 to show that the only WME outcomes are success and failure. Suppose the game arrives at date $\bar{T}$, and the history up to this point features all of the early adopters purchasing tokens, and both supplying and demanding service at each date. By weak monotonicity, late adopters know that the early adopters will continue to behave the same way after any additional quantity of tokens are sold to the late adopters. Then, the sequentially rational action for each late adopter is to purchase a token and do the same. Then, the game will continue to reach the efficient outcome beyond date $\bar{T}$. From this point, we can directly copy the proof of Proposition 2.

Next, we extend Lemma 1 to show that there is a node at date $N+1$ with all early adopters holding tokens that is relevant to the failure outcome. The proof exactly follows the earlier result, using the same equilibrium that was constructed in that proof.

To extend Lemma 3, we invoke the lower bound on $\bar{T}$ in Assumption 1. With this, an early adopter who purchases a token in the sale cannot possibly recoup this utility cost if she waits to participate until the late adopters arrive. Then, any strategy for an early adopter that is weakly monotonic, relevant to the failure outcome, and specifies y at a token-sale node, must specify to

(a) Appendix Proposition 2.

generate the same benefit but impose different costs. The gap in the set of sustainable equilibria suggests that our equilibrium concept, in which users only condition on M , may be too restrictive.

There are many possible ways to enrich the strategy space that give the users the ability (directly or indirectly) to respond differently to the two scenarios described above. Recent trade volume $\hat{M}$ is just one possibility, but it is a particularly intuitive and plausible one. If $P < c+u$, then acquiring on the platform is a “stronger” signal of one’s intent to participate on the platform. When $\hat{M}$ is high, most tokens were recently acquired on the platform for the higher cost. We can then construct an equilibrium of the type we needed for Lemma 1: In this equilibrium, users respond to higher $\hat{M}$ with greater probability of y , and this allows the type A players to be indifferent to paying either cost for a token.

We first revise Definition 6:

Definition 8. A pure strategy for a player i is weakly monotonic if

- (1) its prescribed action at any h depends only on $X_i(h)$, $M(h)$, and $\hat{M}(h)$.
- (2) if it specifies y for an action at any h , then it also specifies y for the same action at any h' where $X_i(h')=X_i(h)$, $M(h')\geq M(h)$, and $\hat{M}(h')\geq \hat{M}(h)$.

We then restore the equilibrium from Lemma 1 by redefining $\alpha = \frac{pw+(1-p)\hat{P}}{p(s-\hat{P})}$, where $\hat{P} \equiv \frac{\hat{M}}{M} \times (c+u) + \left(1 - \frac{\hat{M}}{M}\right) \times P$ is a weighted average of the cost incurred to purchase tokens on the platform or in the token sale. To adapt the rest of the results and proofs, we simply extend any strategies to specify the same action regardless of the value of $\hat{M}$. With this done, the proof of Theorem 1 goes through exactly as stated. We can similarly adapt the analysis of Appendices D and E by simply adjusting the participation probabilities in the equilibria that correspond to Lemma 1 in each section.

There are many alternative signals to $\hat{M}$ that could be used to achieve the same effect. For example, players could condition on whether the prior date was part of the token sale or platform operation. The general point is that $P=c+u$ can be relaxed if we are slightly less strict than in the main model about the information that affects equilibrium strategies.

References

- Adhami, S., G. Giudici, and S. Martinazzi. 2018. Why do businesses go crypto? An empirical analysis of initial coin offerings. *Journal of Economics and Business* 100:64–75.
- Amsden, R., and D. Schweizer. 2018. Are blockchain crowdsales the new “gold rush”? Success determinants of initial coin offerings. Working Paper, Concordia University.
- Aumann, R. 1964. Mixed and behavior strategies in infinite extensive games. In M. Dresher, L. Shapley, and A. Tucker, eds., *Advances in Game Theory*, vol. 52, 627–50. Princeton, NJ: Princeton University Press.

(b) Appendix F: generalized price and Definition 8.

7 Short summary

- The paper provides a theory of digital tokens as coordination devices for launching peer-to-peer platforms.
- A platform with network effects has a success equilibrium and a failure equilibrium.
- A token sale gives users a costly observable way to signal future participation.
- Forward induction makes other users interpret token purchases as evidence of intended platform use.
- In the two-user model, Proposition 1 says the only forward-induction outcome is success.
- In the main N -user repeated platform model, Theorem 1 says all forward-induction outcomes are success outcomes.
- The token works because it is platform-specific, observable, and intrinsically useless outside the platform.
- The model does not require subsidies, commitment to participate, dividends, repurchases, or productive use of sale proceeds.
- With late adopters, investors can buy tokens early and resell them later as access rights.
- With private information, the token auction price aggregates investor signals about platform value.
- The policy discussion suggests that some utility tokens differ economically from conventional securities.
- The welfare conclusion is qualified: tokens improve efficiency when the avoided probability of coordination failure is large relative to any socially wasteful sale cost.

8 Conclusion

8.1 Core conclusion

Li and Mann show that digital tokens can be economically valuable even when they have no dividend, no repurchase promise, and no use outside a platform. Their value is that they allow dispersed potential users to coordinate on platform participation. A token sale is a costly, observable signal of intended future use.

8.2 Economic intuition

The central intuition is forward induction. If a user buys a useless platform token before launch, others should ask why. The rational explanation is that the buyer expects to use the platform. This belief makes others willing to participate, and this in turn makes the original purchase rational. The token sale therefore helps select the good equilibrium.

8.3 Takeaway

The paper reframes utility tokens as platform-building tools rather than financial securities. The key feature of the best coordination token is not that it has many uses, but that it has only one: participation in the platform. That narrow use makes token acquisition informative, and the information solves the platform's coordination failure.